JAVA using BlueJ

Introduction Java

Introduction

- Java is a high-level, class-based, object-oriented programming language that .
- It was first developed by James gosling at Sun Microsystems in the mid-1990s (now owned by Oracle).
- It is widely used for developing a variety of applications, including web, mobile, desktop, and enterprise software.
- Java is known for its simplicity, readability, and platform independence.
- Java was formerly known as Oak.

Java Compilation Process

- An ordinary compiler simply convert the source code (written by a user) into a machine code.
- A machine code is dependent on a specific platform, programming environment or a machine. Hence, it cannot run/execute on every machine such a code is called the ***native code***.
- Unlike other compiler , the java compiler converts the source code into a format, which is neither dependent on a machine nor a platform.
- We call this new code as the **Java byte Code**. Further the byte code is converted into an executable machine code for a particular platform by JVM (java Virtual machine) using the corresponding interpreter.

Java Byte Code

- Like other compilers, java compiler does not create the machine code for any particular platform.
- It produces an intermediate code in a special format, which can be interpreted for different platforms.
- This special format is known as the byte code.
- The same byte code is used by all interpreters to create various type of machine code supporting corresponding platforms.

The Java Virtual Machine (JVM)

- The code written by using any programming language, is either compiled or interpreted before execution. But in Java, we perform both operations.
- First of all, we compile the code into byte code. Further, this byte code is interpreted at the platform level, on which the application is to be executed.
- For every platform, we have a special java interpreter, which interprets the java byte code.
- The devices (computer, mobile, etc.), which are equipped with interpreter that can read the byte code , are called **Java Virtual Machines**.
- Hence any software that has the interpreter to read the **java byte code** can be referred to as a java virtual Machine.

Characteristics of Java

- **Platform Independent** : Java applications are platform-independent, which means that you can write a Java program on one platform and run it on any other platform that supports Java.
- **Object Oriented** : Java follows the object-oriented programming (OOP) paradigm, which means that it uses classes and objects to model real-world concepts and solve problems. Key OOP concepts in Java include inheritance, polymorphism, encapsulation, and abstraction.
- **Robust :** Java has a strong memory management system. It has a unique compilation system, in which the compiler checks the program errors and the interpreter checks any run time errors, and makes the system secure. This two step compilation makes java a robust language.

Characteristics of Java (cont..)

- **Distributed** : Distributed computing involves several computers working together on a network. The widely used protocols like HTTP and FTP are developed in java.
- **Secure :** java uses an encryption system to allow the Java applications to transmit data over the internet in an encrypted format. In this manner; It makes data transmission on secure, using the java applications.
- **Multithread :** Java is also a multithreaded programming language. Multithreading works in similar way as multiple processes run on one computer.
- **Support multimedia** : Java application support multimedia content, i.e., graphics, animation, image and video.
- **Freely Available** : It can be downloaded and distributed freely.

Question Answer

- 1. What was Java initially called?

  a Oak

  b. C

  c. C++

  d. None of these

- 2. What is Java Programming Language?

  a. A runtime system

  b. A set of development tools.

  c. An Application Programming Interface (API)

  d. All of these

- 3. Name the process that converts source code to bytecode.

  a. Interpretation

  b. Compilation

  c. All of these

  d. None of these

- 4. A Virtual Processor that is implemented in software and runs using the capabilities provided by your operating system and computer hardware.

  a. Byte Code

  b. Compiler

  c. Interpreter

  d. Java Virtual Machine(JVM)

Question Answer

- Which among the following is not a language feature in Java?
  - a. Robust
  - b. Secured
  - c. Platform Independent
  - d. Procedure Oriented
- Name the programs that can be developed in such a way that it remains embedded in a web page and runs on the viewer's machine in a secured manner by Java compatible browsers.
  - a. Applets
  - b. Applications
  - c. Both a and b
  - d. None of these
- Name the Application program that is written and compiled which may then be executed in any machine provided it contains the JVM.
  - a. Applets
  - b. Applications
  - c. Both a and b
  - d. None of these
- What is the extension of a source code in Java?
  - a. .java
  - b. .class
  - c. Both a and b
  - d. None of these

Question Answer

- What is the extension of the byte code in Java?
  - a. .java
  - b. .class
  - c. Both a and b
  - d. None of these
- What is a set of pseudo machine language instructions that are understood by the Java Virtual Machine and are independent of the underlying hardware called?
  - a. JVM
  - b. Source Code
  - c. Compilation
  - d. Bytecode